# EXP 6: WORKFLOW AUTOMATION USING DELUGE IN ZOHO CREATOR

## AIM

To automate the Library Book Issue Management application in Zoho Creator using Deluge scripting, with form workflows, scheduled workflows, batch workflows, custom functions and blueprints.

## PROCEDURE

1. Open the application, click **Edit this application** and go to the **Workflow** tab.
2. **Form workflow:** create **On_IssueRequest_Submit_Workflow** on the Book Issue Request form (Created or Edited, on successful form submission) and write the Deluge actions.
3. **Email action:** create **Send_Book_Issued_Notification** to send a confirmation email to the member on form submission.
4. **Scheduled workflow:** under Schedules, create **Daily_Overdue_Reminder** to run daily at 09:00:00 and email collection reminders.
5. **Custom function:** under Functions, create the Deluge function **calculate_Member_Total_Fine** (return type float) that totals the fine for a member's overdue books, then execute it to test.
6. **Batch workflow:** under Batch workflows, create **Batch_Overdue_Fine_Calculation** to calculate the fine amount for records (overdue days x daily rate), then execute it with a Record ID.
7. **Blueprint:** create **Issue_Lifecycle_Blueprint** with stages Requested, Approved, Book Issued and Book Returned, and transitions Approve Request, Issue Book and Return Book.
8. Save and enable all workflows, submit a test record, and verify the emails and record values.

## OUTPUT SCREENSHOTS

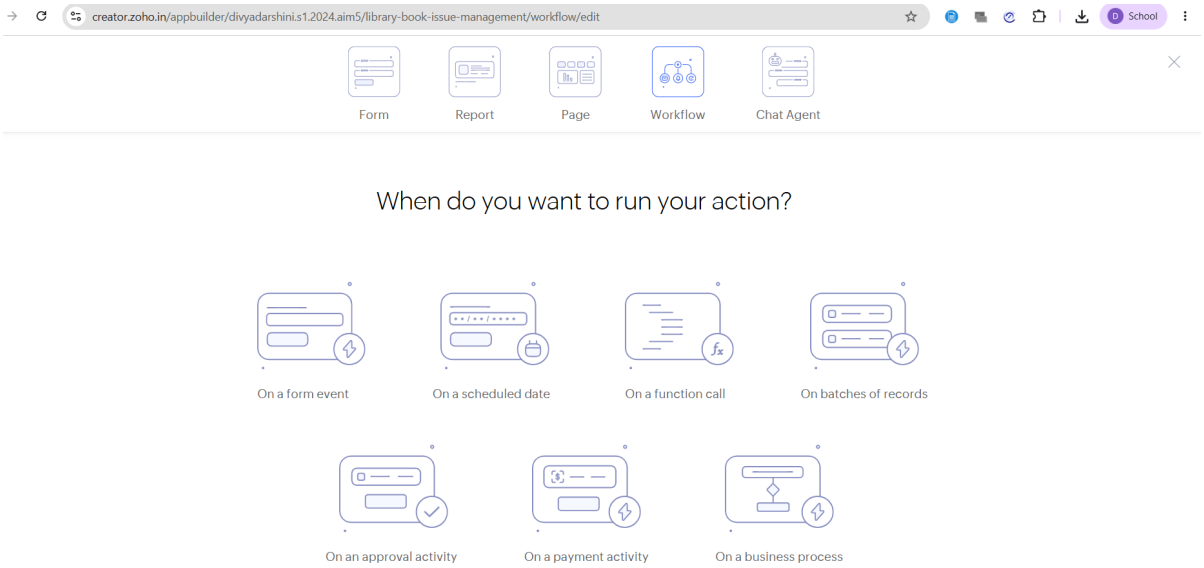



Fig 6.1: Creating a workflow - selecting when to run the action

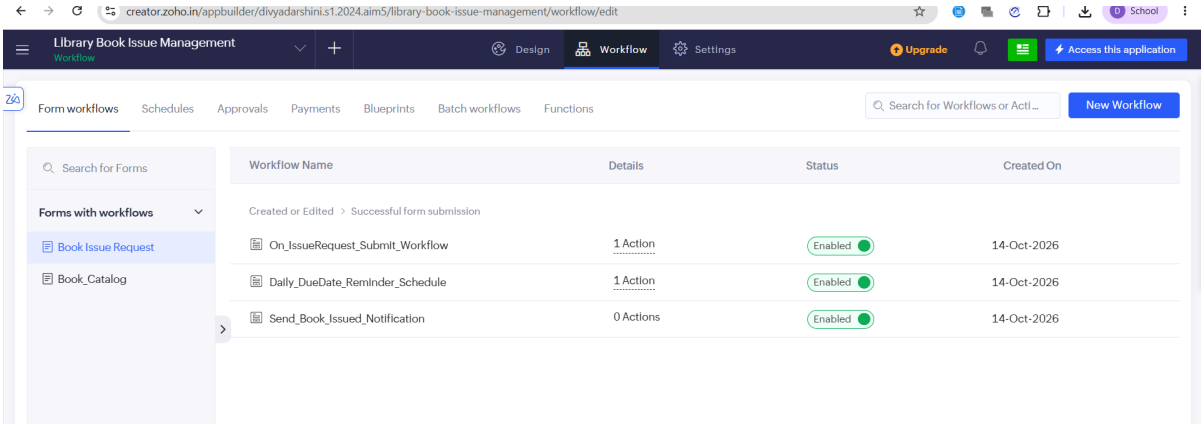



Fig 6.2: Form workflows - On_IssueRequest_Submit_Workflow and Send_Book_Issued_Notification

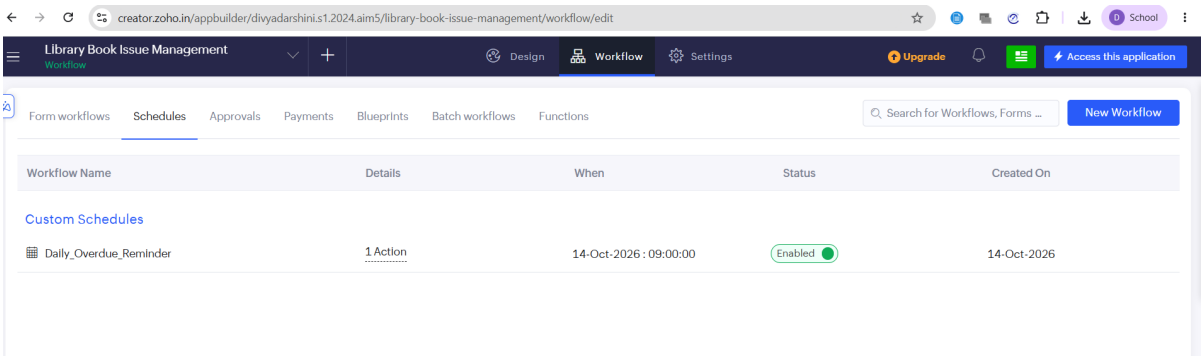



Fig 6.3: Schedules - Daily_Overdue_Reminder (scheduled workflow, runs at 09:00:00)

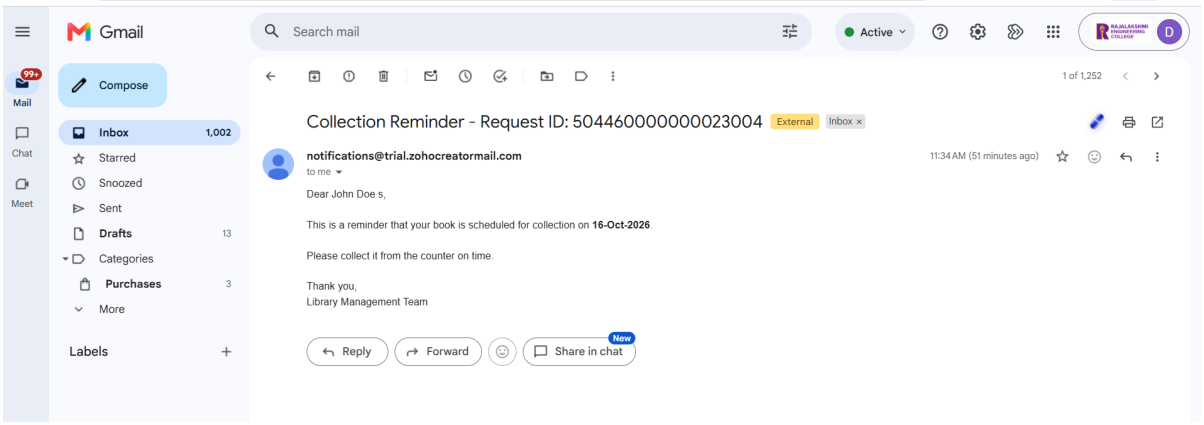



Fig 6.4: Output - Collection Reminder email received in Gmail

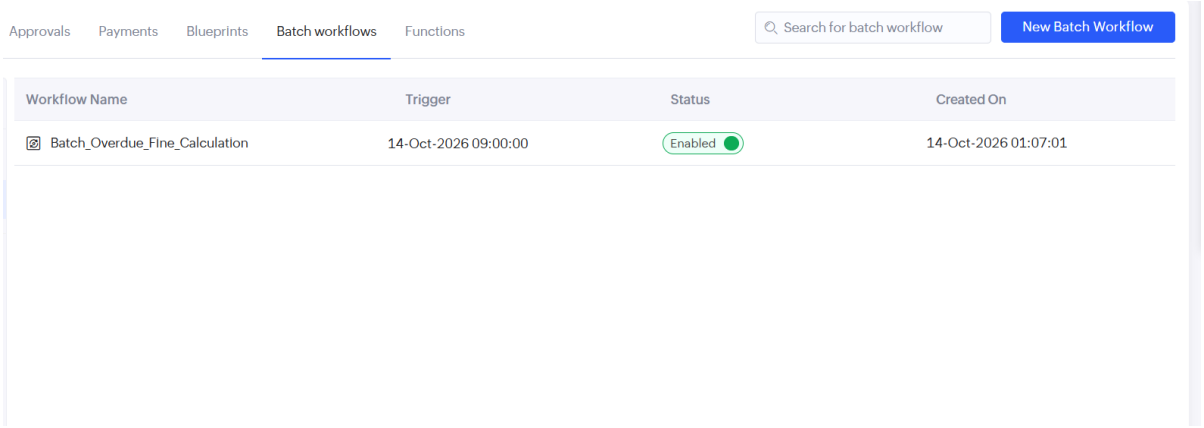



Fig 6.5: Batch workflows - Batch_Overdue_Fine_Calculation




Fig 6.6: Executing the batch workflow with a Record ID

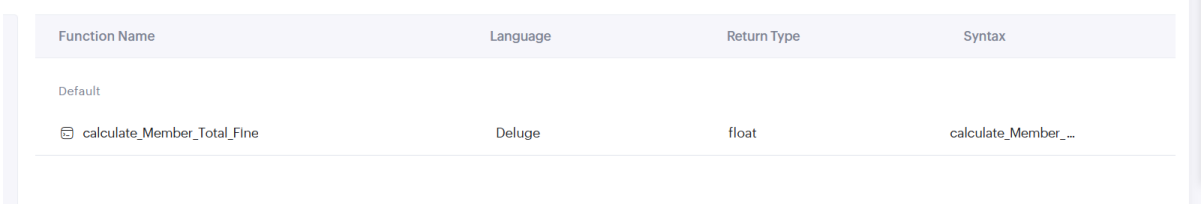



Fig 6.7: Functions - calculate_Member_Total_Fine (Deluge, return type float)

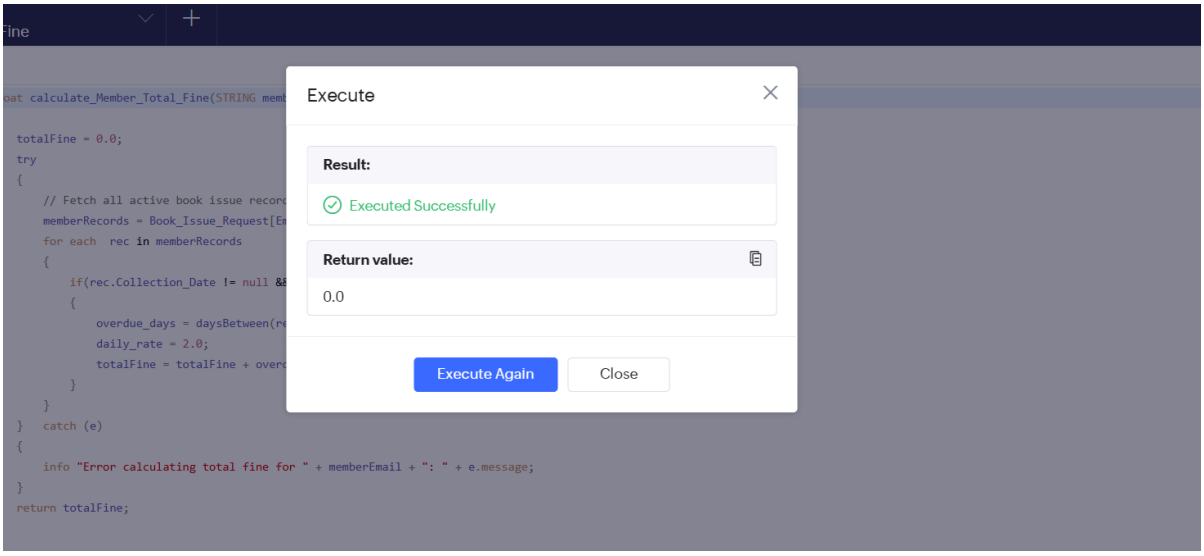



Fig 6.8: Custom function executed successfully

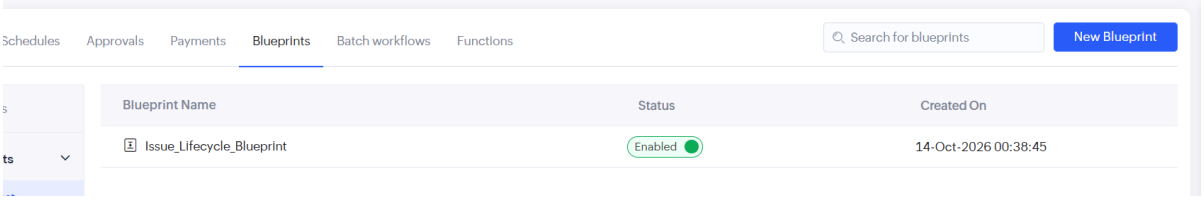



Fig 6.9: Blueprints - Issue_Lifecycle_Blueprint (Enabled)

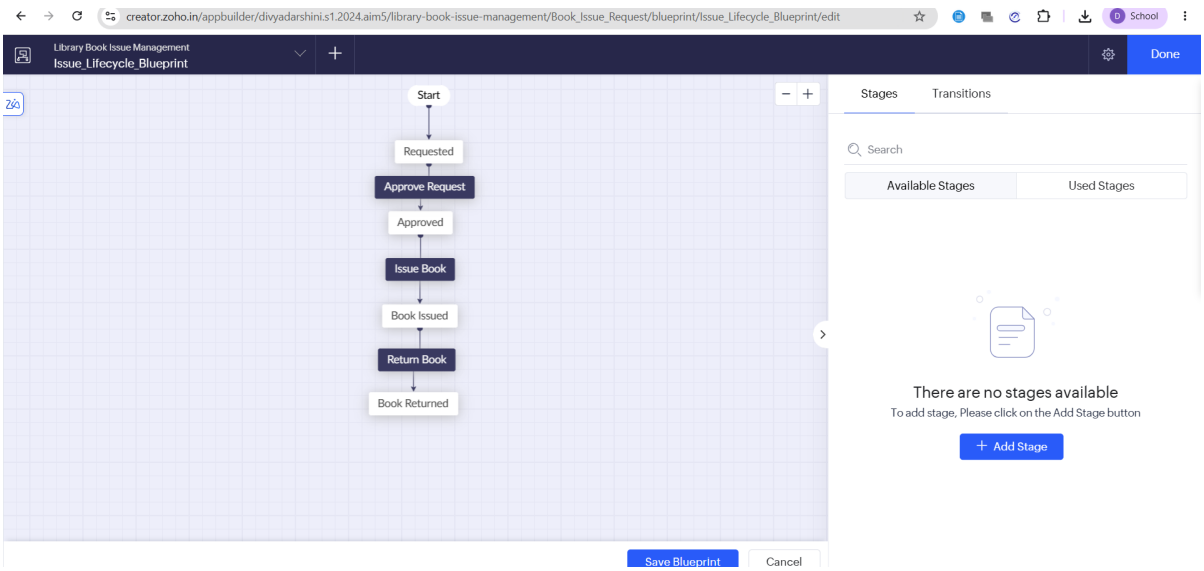



Fig 6.10: Issue_Lifecycle_Blueprint stages: Requested to Book Returned

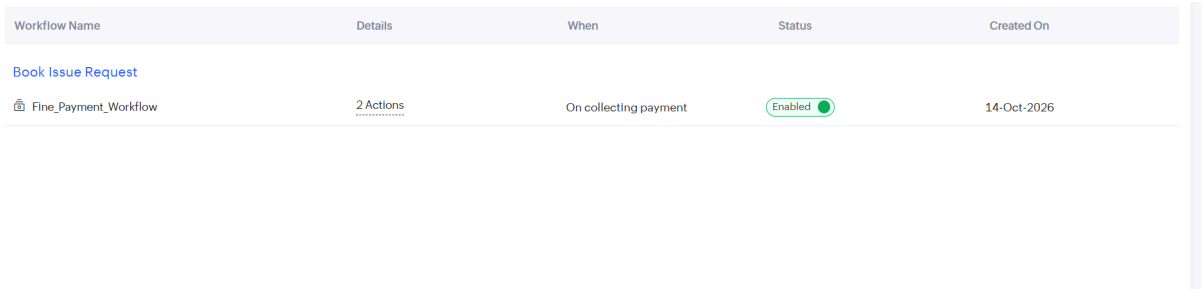



Fig 6.11: Payments - Fine_Payment_Workflow (On collecting payment)

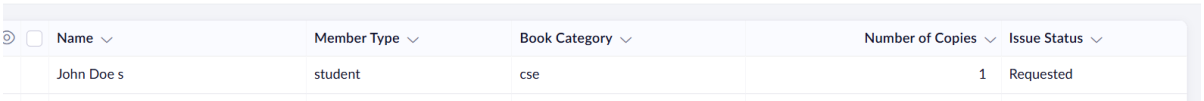



Fig 6.12: Output - Book Issue Request record with Issue Status

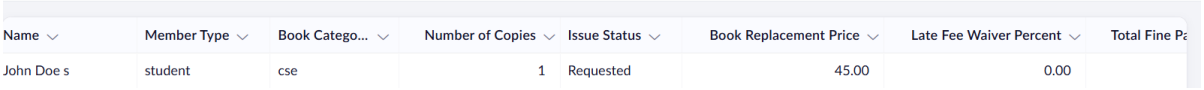



Fig 6.13: Output - Record showing Book Replacement Price and Late Fee Waiver Percent

## RESULT

Workflow automation using Deluge was successfully implemented. The form workflow, scheduled workflow, email notification, batch workflow, custom function and blueprint were created and enabled, and the reminder email and fine calculation worked as expected.